

LESSON 6-1 • SESSION 2 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Multiplication as Scaling

Builds on Grade 5 • 5.NF.B.5

TODAY'S LESSON

Priya says $\frac{3}{4}$ yard cut into 4 pieces makes 3-yard pieces. You will decide if that makes sense.

YOU NEED TO

Tell if multiplying by a fraction makes a number bigger or smaller. **You will use it in Practice Form A, problem 5.**

STEPS TO FOLLOW*use these same steps on every problem*

- 1** Look at the **fraction** you multiply by.
- 2** **Compare** its top and bottom. A smaller top means less than 1.
- 3** Less than 1 gives a **smaller** answer. More than 1 gives bigger.

WATCH ME*the steps in action***Is $\frac{3}{4} \times \frac{1}{4}$ bigger or smaller than $\frac{3}{4}$?**

- Step 1** The fraction is $\frac{1}{4}$.
- Step 2** 1 is less than 4, so $\frac{1}{4}$ is less than 1.
- Step 3** So $\frac{3}{4} \times \frac{1}{4}$ is smaller than $\frac{3}{4}$.

Answer Smaller. (It is $\frac{3}{16}$.)**TRY IT WITH HELP***fill in each step*

- 1.** Is $8 \times \frac{2}{3}$ bigger or smaller than 8?
Step 1 The fraction is _____.
Step 2 2 is less than _____, so $\frac{2}{3}$ is less than 1.
Step 3 So $8 \times \frac{2}{3}$ is _____ than 8.
- 2.** Is $6 \times \frac{5}{4}$ bigger or smaller than 6?
Step 1 The fraction is _____.
Step 2 5 is _____ than 4, so $\frac{5}{4}$ is more than 1.
Step 3 So $6 \times \frac{5}{4}$ is _____ than 6.

YOUR TURN*use the steps on your own*

- 3.** Is the product bigger or smaller than the first number?

- | | | |
|---|--|---|
| a. $10 \times \frac{1}{2}$ _____ | b. $4 \times \frac{3}{2}$ _____ | c. $9 \times \frac{2}{3}$ _____ |
| d. $5 \times \frac{7}{4}$ _____ | e. $\frac{1}{2} \times \frac{1}{3}$ _____ | f. $12 \times \frac{1}{6}$ _____ |

Divide by a Unit Fraction

Builds on Lesson 6-1 Session 1 • 6.NS.A.1

TODAY'S LESSONYou will find $3/4 \div 4$ with the same Keep, Change, Flip steps from Session 1.**YOU NEED TO**Divide by a unit fraction with Keep, Change, Flip, like $6 \div 1/3$. **You will use it in Practice Form A, problems 2 and 5.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Keep** the first number. Write a whole number over 1.
- 2 Change** $\div$ to $\times$ and flip the second number.
- 3 Multiply.** Then simplify.

WATCH ME*the steps in action***Find $6 \div 1/3$.****Step 1** Keep $6 = 6/1$.**Step 2** Change to $\times$. Flip $1/3$ to $3/1$.**Step 3** $6/1 \times 3/1 = 18$.**Answer 18****TRY IT WITH HELP***fill in each step***4.** Find $5 \div 1/4$.**Step 1** Keep $5 = \underline{\hspace{2cm}}/1$ **Step 2** Change to $\times$. Flip $1/4$ to $\underline{\hspace{2cm}}/1$ **Step 3** $5 \times 4 = \underline{\hspace{2cm}}$ **5.** Find $3 \div 1/8$.**Step 1** Keep $3 = \underline{\hspace{2cm}}/1$ **Step 2** Change to $\times$. Flip $1/8$ to $\underline{\hspace{2cm}}/1$ **Step 3** $3 \times 8 = \underline{\hspace{2cm}}$ **YOUR TURN***use the steps on your own***6.** Divide. Use Keep, Change, Flip.

a. $2 \div 1/5 = \underline{\hspace{2cm}}$

b. $7 \div 1/2 = \underline{\hspace{2cm}}$

c. $4 \div 1/6 = \underline{\hspace{2cm}}$

d. $9 \div 1/3 = \underline{\hspace{2cm}}$

e. $10 \div 1/4 = \underline{\hspace{2cm}}$

f. $6 \div 1/10 = \underline{\hspace{2cm}}$

READY CHECK*a problem like the one in today's lesson***7.** Find $3/4 \div 4$. Use Keep, Change, Flip.**A** 3**B** $3/16$ **C** $16/3$ **D** $1/3$ **CHECK YOURSELF**☐ I can do this on my own☐ I need a little help☐ I have a question about #